MIL-R-51472A(EA)
9 October 1986
SUPERSEDING
MIL-R-51472(EA)
4 June 1986

MILITARY SPECIFICATION

REFILL KIT, M58A1 PERSONAL DECONTAMINATION KIT TRAINING AID

This specification is approved for use within Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of refill kit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-I-735 - Isopropyl Alcohol
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPD-TS, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Ammunition Lot Numbering

2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

5-77-2267 - Reagent Water
5-77-2367 - Training Aid, Personal Decontamination Kit, M58A1
5-77-2381 - Decon 1 Packet Training
5-77-2382 - Decon 2 Packet Training

PUBLICATIONS

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

SPECIAL PACKAGING INSTRUCTIONS (SPI)

PMIL-R-51472 - Refill Kit, M58A1 Personal Decontamination Kit Training Aid

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components. The refill kit shall consist of 30 decon 1 packets and 30 decon 2 packets in a fiberboard box.

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3.1.1 Decon 1 packets. The decon 1 packets used in the refill kit shall conform to Drawing 5-77-2381.

3.1.2 Decon 2 packets. The decon 2 packets used in the refill kit shall conform to Drawing 5-77-2382.

3.1.3 Fiberboard box. The fiberboard box used in the refill kit shall conform to style RSC, minimum grade W6c, type CF of PPP-B-636 and shall have inside dimensions of 219 by 143 by 108 millimeters (mm). The box shall contain 10 cells arranged in a two by five cell pattern. The cells shall be formed by interlocking partitions fabricated from fiberboard conforming to style SW, type CF, grade 125, class domestic of PPP-F-320.

3.2 Assembly. Each cell in the fiberboard box shall contain three decon 1 packets and three decon 2 packets arranged in the same manner as the packets in the M58A1 training aid shown on Drawing 5-77-2367. The filled box shall be closed as specified for interior containers in the appendix to PPP-B-636.

3.3 Ampoule leakage. The ampoules shall not leak when tested as specified in 4.4.4.1.

3.4 Ampoule fill. Each ampoule shall be filled to a height of 43 mm to 51 mm with a solution consisting of 50 ± 2 percent by weight 2-propanol conforming to TT-I-735, with the remainder water conforming to Drawing 5-77-2267 when tested as specified in 4.4.4.2. The contractor shall certify that the composition of the solution meets the requirements of this specification.

3.5 Decon 1 packet leakage. The decon 1 packet shall not leak when tested as specified in 4.4.4.3.

3.6 Decon 2 packet leakage. The decon 2 packet shall not leak when tested as specified in 4.4.4.3.

3.7 Decon 1 packet solution content. The pad in the decon 1 packet shall contain 3.4 to 4.9 grams (g) of a solution consisting of 50 ± 2 percent by weight 2-propanol conforming to TT-I-735, with the remainder water conforming to Drawing 5-77-2267 when tested as specified in 4.4.4.4. The contractor shall certify that the composition of the solution meets the requirements of this specification.

3.8 Marking. The fiberboard box shall be marked in accordance with 5.1.

3.9 Workmanship. The refill kit and all components and subassemblies thereof shall be free from defects and contamination by foreign matter such as dirt, grease, and oil.

3.10 Preproduction. Prior to the start of regular production, a preproduction sample of refill kits shall be produced in accordance with this specification for examination and test (see 4.3).

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3.10.1 Accelerated aging. The refill kit shall meet all the test requirements in this specification after accelerated aging as specified in 4.3.3.1.

3.10.2 Cold storage stability. The refill kit shall meet all the test requirements in this specification after cold storage as specified in 4.3.3.2.

3.11 Pad sticking. During removal, the pad (towelette) shall not stick to the packet when tested as specified in 4.4.4.5.

3.12 Ampoule tip. The sealed end of the ampoule shall be rounded and smooth to the touch when tested as specified in 4.4.4.6.

3.13 Ampoule screen. The screen shall show no evidence of damage or separation in the heat seal area when tested as specified in 4.4.4.7.

3.14 Packet components. There shall be no missing or damaged Decon I and Decon II packet components when tested as specified in 4.4.4.8.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

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4.3 Preproduction inspection.

4.3.1 Lotting. A preproduction lot of no less than 500 refill kits shall be produced using the same methods, materials, equipment, and processes of manufacture as will be used during regular production.

4.3.2 Sampling and inspection procedure.

4.3.2.1 For examination. Sampling and inspection procedure for examination shall be as specified in 4.4.2.1 and 4.4.3.1.

4.3.2.2 For test. Sampling and inspection procedure for test shall be as specified in table I.

TABLE I. Preproduction inspection

Requirement	Sample size	Accept/reject criteria	Test paragraph
Accelerated aging (3.10.1)	1 refill kit		4.3.3.1
Cold storage stability (3.10.2)	1 refill kit		4.3.3.2
Tests after accelerated aging or cold storage:			
Ampoule leakage (3.3)	90	Acc 3 Rej 4	4.4.4.1
Ampoule fill (3.4)	30	Acc 2 Rej 3	4.4.4.2
Decon 1 packet leakage (3.5)	30	Acc 1 Rej 2	4.4.4.3
Decon 2 packet leakage (3.6)	30	Acc 1 Rej 2	4.4.4.3
Decon 1 packet solution content (3.7)	10	Acc 0 Rej 1	4.4.4.4
Pad sticking (3.11)	30* Decon I packets		
	30** Decon II packets	Acc 0 Rej 1	4.4.4.5
Ampoule tip (3.12)	90***	Acc 0 Rej 1	4.4.4.6
Ampoule screen (3.13)	30** Decon II packets	Acc 1 Rej 2	4.4.4.7
Packet components (3.14)	30* Decon I packets		
	30** Decon II packets	Acc 0 Rej 1	4.4.4.8

*Same sample packets used for Decon I leakage test may be used.
**Same sample packets used for Decon II leakage test may be used.
***Same sample packets used for ampoule leakage test may be used.

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4.3.3 Tests. Tests shall be conducted as follows:

4.3.3.1 Accelerated aging. Place the refill kits in an oven at 71° ± 1°C for 24 hours. Remove the refill kits from the oven and allow to reach ambient temperature.

4.3.3.2 Cold storage stability. Store the refill kits at -54° ± 1°C for 24 hours. Remove the refill kits from cold storage and allow to reach ambient temperature.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot of refill kits or components shall consist of the refill kits or components produced by one manufacturer, at one plant, under essentially the same manufacturing conditions. However, no more than one interfix number of decon I packets or decon II packets shall be represented in any one lot of kits. Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For test. Sampling shall be conducted in accordance with table II.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample refill kits and components shall be examined in accordance with MIL-STD-105 and the following classification of defects.

(a) Refill Kit, M58A1 Personal Decontamination Kit Training Aid.

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective for defects grouped collectively
101	Components missing or incorrect (see 3.1)
102	Fiberboard box not as specified (see 3.1.3)
103	Assembly incorrect (see 3.2)
104	Marking incorrect, illegible, or missing (see 3.8)
105	Components damaged
106	Foreign matter in or on refill kit (see 3.9)

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TABLE II. Quality conformance inspection

Requirement	Sample size	Accept/reject criteria	Test paragraph
Ampoule leakage (3.3)	MIL-STD-105, inspection level S-4, prior to assembly in packets	AQL 2.5 percent defective	4.4.4.1
Ampoule fill (3.4)	MIL-STD-105, inspection level S-3, prior to assembly in packets	AQL 2.5 percent defective	4.4.4.2
Decon 1 packet leakage (3.5)	MIL-STD-105, inspection level S-3, prior to assembly in refill kits	AQL 4.0 percent defective	4.4.4.3
Decon 2 packet leakage (3.6)	MIL-STD-105, inspection level S-3, prior to assembly in refill kits	AQL 4.0 percent defective	4.4.4.3
Decon 1 packet solution content (3.7)	15, prior to assembly in refill kits	Acc 1 Rej 2	4.4.4.4
Pad sticking (3.11)			
Decon I packet	MIL-STD-105, inspection level S-3, prior to assembly*	AQL 4.0 percent defective	4.4.4.5
Decon II packet	MIL-STD-105, inspection level S-3, prior to assembly**	AQL 4.0 percent defective	4.4.4.5
Ampoule tip (3.12)	MIL-STD-105, inspection level S-4, prior to assembly***	AQL 4.0 percent defective	4.4.4.6
Ampoule screen (3.13)	MIL-STD-105, inspection level S-3, prior to assembly**	AQL 4.0 percent defective	4.4.4.7
Packet components (3.14)			
Decon I packet	MIL-STD-105, inspection level S-3, prior to assembly*	AQL 4.0 percent defective	4.4.4.8
Decon II packet	MIL-STD-105, inspection level S-3, prior to assembly**	AQL 4.0 percent defective	4.4.4.8

*Same sample packets used for Decon I leakage test may be used.
**Same sample packets used for Decon II leakage test may be used.
***Same sample packets used for ampoule leakage test may be used.

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(b) Decon 1 Packet Training and Decon 2 Packet Training (Drawings 5-77-2381, 5-77-2382, and subsidiary drawings). The following defects may be examined prior to refill kit assembly:

Categories		Defects
Critical:		None defined
Major:	AQL 2.5 percent defective for defects grouped collectively	
	101	Packet heat seal width incorrect
	102	Packet notch dimensions incorrect
	103	Packet notch locations incorrect
	104	Packet notches not cleanly cut
	105	Packet marking incorrect, illegible, or missing

(c) Packaging inspection (section 5).

Categories		Defects
Critical:		None defined
Major:	AQL 4.0 percent defective for defects grouped collectively	
	101	Container not as specified
	102	Contents of container not as specified
	103	Container closure not as specified
	104	Palletization incorrect
	105	Marking missing, incorrect, or illegible
	106	Shelf life marking missing, incorrect, or illegible
	107	Lot number missing, incorrect or illegible

4.4.3.2 For test. Sample refill kits shall be tested as specified in table II.

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Ampoule leakage. Place the ampoules in a vacuum tank and weight with a suitable cover to ensure complete submersion when the tank is filled with water. Place sufficient water soluble dye in the water so that one drop of the solution will impart visible color to the solution in the ampoule. Reduce the pressure in the tank to 127 mm of mercury or less absolute, maintain at that pressure for at least 25 minutes, and then pressurize to ambient. After no less than 20 minutes at ambient pressure, remove the ampoules from the tank, rinse thoroughly in clean water, and spread on a drying screen. Examine the dry capsules against a white background. Colored solution in an ampoule shall be considered evidence of leakage.

4.4.4.2 Ampoule fill. With the ampoule held vertically on a flat surface, measure the height of the meniscus above the surface to the nearest millimeter.

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4.4.4.3 Packet leakage. Submerge the packet for at least 60 seconds in water heated to 58.8° ± 1.5°C. Keep the packet submerged in a vertical position with the uppermost surface covered by no more than 25 mm of water. Continuously examine the packet during the time it is submerged for evidence of bubbles. The presence of bubbles shall be considered evidence of leakage.

4.4.4.4 Decon 1 packet solution content. Remove the pad from the packet and weigh to the nearest 0.01 g. Place the pad in an air circulating oven and dry at 105° ± 2°C for 60 ± 5 minutes. Weigh to the nearest 0.01 g. Calculate the weight of solution on the pad as follows:

$$\text{Weight of solution} = A - B$$

where: A = Weight of pad before drying in grams, and
B = Weight of pad after drying in grams.

4.4.4.5 Pad sticking. Open the packet and remove the pad (towelette).

4.4.4.6 Ampoule tip. Hold the ampoule and touch the sealed end. Inspect for evidence of a pointed tip.

4.4.4.7 Ampoule screen. Hold the screen and observe for compliance to the specified requirement.

4.4.4.8 Packet components. Empty the packets and visually inspect the components.

5. PACKAGING

5.1 Packaging. Packaging shall be as specified in SPI PMIL-R-51472.

6. NOTES

6.1 Intended use. The refill kit covered by this specification is intended for use in refilling the Training Aid, Personal Decontamination Kit, M58A1 with decon 1 packets and decon 2 packets.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of packing required (see 5.1)
(c) Shelf life code
(d) Palletization required

6.3 Subject term (key word) listing.

Decontamination
Decontamination kits

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-51472A(EA)	REFILL KIT, M58A1 PERSONAL DECONTAMINATION KIT TRAINING AID

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include A Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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